Artificial Intelligence - MCQ - Slot C

Total points 19/20

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CO3-AT2

In the process of Knowledge Engineering for a domain, what step immediately follows **assembling the relevant knowledge**? *1/1

- ○ Encoding general axioms about the domain.
- ◉ Deciding on a vocabulary of predicates, functions, and constants. ✓
- ○ Positing specific queries to the inference engine.
- ○ Skolemizing the KB.

✓ When Skolemizing a sentence where an existential quantifier lies within the scope of universal quantifiers (e.g., ∀x ∃y HasChild(x, y)), the existential variable y must be replaced by: * 1/1

- ○ A unique Skolem constant (e.g., C_1)
- ◉ A Skolem function of the universally quantified variable (e.g., f(x)) ✓
- ○ A free variable y
- ○ The variable x

✓ What is **Generalized Modus Ponens (GMP)**? * 1/1

- ○ An inference rule for propositional resolution.
- ◉ A lifted inference rule that combines Unification with Modus Ponens directly on quantified Horn clauses. ✓
- ○ A rule that eliminates existential quantifiers using Skolemization.
- ○ An algorithm for converting FOL to CNF.

✓ What is the result of resolving Clause 1: ¬P(x) ∨ Q(x) with Clause 2: P(John) ∨ R(Jane)? * 1/1

- ◉ Q(John) ∨ R(Jane) ✓
- ○ ¬P(John) ∨ R(Jane)
- ○ Q(x) ∨ R(Jane)
- ○ ∅ (Empty Clause)

Given two terms p = Knows(John, x) and q = Knows(y, Mother(y)), what is their Most General Unifier (MGU)? *1/1

- [x] {x/Mother(John), y/John} ✓
- [ ] {x/John, y/Mother(y)}
- [ ] {x/Jane, y/John}
- [ ] Unification Fails

Which of the following statements about **quantifier scope and order** in FOL is TRUE? *1/1

- [ ] ∀x ∃y Loves(x, y) is logically equivalent to ∃y ∀x Loves(x, y).
- [ ] Switching the order of two universal quantifiers (∀x ∀y) changes the meaning of the statement.
- [x] ∃y ∀x Loves(x, y) implies ∀x ∃y Loves(x, y), but the converse is not generally true. ✓
- [ ] Quantifiers can never be nested inside predicates.

What is **Factoring** in First-Order Resolution, and why is it necessary for completeness? *1/1

- [x] Factoring combines two unifiable literals within the same clause into a single literal using MGU θ. ✓
- [ ] Factoring splits a clause into two smaller clauses.
- [ ] Factoring removes all function symbols.
- [ ] Factoring converts disjunctions to implications.

✓ Consider the propositional sentence α = $(P \lor Q) \land (\neg P \lor R)$. Under which truth assignment for (P, Q, R) does α evaluate to False? * 1/1

- ( ) (T, T, T)
- ( ) (F, T, T)
- (•) (T, F, F) ✓
- ( ) (F, T, F)

✓ A propositional formula is in **Conjunctive Normal Form (CNF)** if it is: * 1/1

- (•) A conjunction of clauses, where each clause is a disjunction of literals. ✓
- ( ) A disjunction of clauses, where each clause is a conjunction of literals.
- ( ) A series of nested implication statements.
- ( ) A single negated implication.

✓ Which of the following sentences correctly represents "There is exactly one king" in FOL? * 1/1

- ( ) $\exists x\ (King(x))$
- (•) $\exists x\ (King(x) \land \forall y\ (King(y) \Rightarrow x = y))$ ✓
- ( ) $\forall x\ (King(x) \Rightarrow \exists y\ (x = y))$
- ( ) $\exists x\ \forall y\ (King(x) \lor King(y))$

The process of **Propositionalization** converts a FOL KB into Propositional Logic by creating instances of quantified sentences. What is a major theoretical limitation of propositionalization? *1/1

- It is unsound.
- If the KB contains function symbols, the set of ground instances becomes infinite. ✓
- It cannot handle implications.
- It converts Horn clauses to non-Horn clauses.

Why does unification fail for the terms p = Knows(John, x) and q = Knows(x, Jack)? *1/1

- John and Jack are both functions.
- Variable x cannot simultaneously bind to two different constants (John and Jack). ✓
- The terms have different predicate arities.
- Predicates cannot take variables as arguments.

Which of the following components are part of **First-Order Logic (FOL)** but are **NOT** present in Propositional Logic? *1/1

- Logical connectives (∧, ∨, ¬)
- Objects, Relations (Predicates), and Quantifiers (∀, ∃) ✓
- Truth values (True / False)
- Sentences and Parentheses

To prove that KB $\models \alpha$ using Resolution Refutation: * 1/1

- ○ Add α to the KB and derive a tautology.
- ◉ Add $\neg\alpha$ to the KB, convert all clauses to CNF, and derive the empty clause ($\emptyset$). ✓
- ○ Show that KB $\wedge\ \alpha$ is satisfiable.
- ○ Show that KB $\Rightarrow \neg\alpha$ is true.

The First-Order Resolution Rule resolves two clauses: $l_1 \vee \ldots \vee l_k$ and $m_1 \vee \ldots \vee m_n$ if: * 1/1

- ○ l_i and m_j are identical.
- ◉ l_i and $\neg m_j$ are unifiable with MGU θ. ✓
- ○ Both clauses contain only ground terms.
- ○ Both clauses are Horn clauses.

In Knowledge Engineering, what is the process of **Ontological Engineering**? * 1/1

- ○ Writing efficient Python algorithms for inference.
- ◉ Representing abstract general concepts—such as Actions, Time, Physical Objects, and Beliefs—in a structured hierarchy. ✓
- ○ Converting all First-Order formulas to C code.
- ○ Removing all quantifiers from a knowledge base.

✓ Propositional resolution is **sound** and **refutation complete**. What does "refutation completeness" mean? *1/1

- ○ Resolution can derive every sentence that is entailed by the KB.
- ◉ If a set of clauses is unsatisfiable, resolution will always be able to derive a contradiction (the empty clause $\emptyset$). ✓
- ○ Resolution runs in polynomial time for arbitrary propositional formulas.
- ○ Resolution never generates redundant clauses.

✓ Consider the assertion: "No two people have the same social security number (SSN)." Select the correct FOL formulation: *1/1

- ○ $\forall x\ \forall y\ (SSN(x) = SSN(y) \Rightarrow x = y)$
- ○ $\forall x\ \forall y\ (x \neq y \Rightarrow SSN(x) \neq SSN(y))$
- ◉ Both A and B are correct. ✓
- ○ Neither A nor B is correct.

✗ Two logical sentences α and β are logically equivalent ($\alpha \equiv \beta$) if and only if: *0/1

- ○ $\alpha \models \beta$
- ◉ $\alpha \Rightarrow \beta$ is a tautology ✗
- ○ $\alpha \models \beta$ and $\beta \models \alpha$
- ○ $\alpha \land \beta$ is satisfiable

Correct answer

- ◉ $\alpha \models \beta$ and $\beta \models \alpha$

Which of the following properties distinguishes a **logical agent** using Knowledge-Based Reasoning from a simple reflex agent? *1/1

- ○ Logical agents map precepts directly to actions via lookup tables.
- ◉ Logical agents maintain an internal state (Knowledge Base) represented in formal logic to deduce hidden aspects of the world. ✓
- ○ Logical agents only use probabilistic models to make decisions.
- ○ Logical agents do not require a domain-specific inference engine.